

Physical Experiment *II*

Lab Report

Lab Title: The Franck-Hertz Experiment

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Instructor: 付浩

Teaching Assistant: 李博洋

Date Performed: 2023.10.19

Score:

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Abstract (About 100 words, 3 points)

The aim of this experiment is to Determine the first excitation potential of the argon atom and promote our understanding of the process of electron collision with an atom And the process of electron collision with an atom. Measurement were made by changing the voltage between the tubes filled with argon atoms and observing the outcome on the multimeter. The voltage difference between two crests were the first excitation In my experiment, the first excitation potential of argon was 11.6V.

Score

Introduction (3 points)

The Franck-Hertz experiment was carried out in1914 by James Franck and Gustav Hertz. Through this experiment, they discovered that electrons were accelerated and pass through mercury vapor, where they lost energy by inelastic scattering in quantized steps as they excited mercury atoms from the ground state to an excited state. This discovery proved the theory of Bohr model and Determine the first excitation potential of the atom. James Franck and Gustav Hertz were awarded Nobel prize because of this..

In this experiment, the physical principles mainly used are the theories of elastic collision and the inelastic collisions as well as the Bohr's Model to address the problem. We used the Franck-Hertz apparatus to create the diagram of the voltage and current with the help of the computers to determine the first excitation potential of the argon atom.

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Experimental Procedure (State main steps in order of

performance, 3 points)

1. Connect the wires and turn on the computer. Preheat the apparatus.
2. Find the recommended values for the tube and pour apparatus, which are V_1 V_2 V_3 V_4
3. Start the software. Set the voltages of V1, V2, V3, V4 with the recommended values. Check these voltages again. Run the program and observe a curve on the scree. The recommended voltages of V1, V2, V3, V4 are only acceptable if more than four crests or troughs would be observed on the curve. If the crests or troughs are less than four, increase Vi (or V2) or decrease Vs in intervals of 0.1 volt.
4. Measure the current (IA)-accelerating voltage (V4) relationship by manual mode. Set the values of V1, V2, V3 on the panel. Increase V4 in intervals of 1V and read the current from the picoammeter and enter the results in Data Table
5. Measure the current (IA)-accelerating voltage (V4) relationship by computer.
6. Maintain the voltages of V2 and Vs as in Data Table 3.10-1 and increase the value of Vi in step of 0.1 V (or 0.2 V) from its initial value in previous Data Table and enter the data

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Results (Data tables and figures, 2 points)

1. Data Tables

DATA TABLE 3-1 (Measured by manual mode. *purpose*: to determine the first excitation potential of argon atom)

V_1= *2.0V* ; V_2= *2.0V* ; V_3= *8.0V* ;

(The unit of the current in the following table is 10^{-7} A)

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	0.0	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0
0	0	0	0	0	0	0	0	0	0	0
10	0.001	0.052	0.185	0.346	0.509	0.463	0.583	0.659	0.671	0.622
20	0.537	0.448	0.366	0.311	0.369	0.663	1.104	1.468	1.716	1.758
30	1.612	1.297	0.918	0.595	0.418	0.435	0.755	1.320	1.910	2.328
40	2.547	2.504	2.212	1.725	1.191	0.767	0.570	0.687	1.149	1.760
50	2.324	2.748	2.961	2.928	2.666	2.237	1.723	1.263	0.967	1.002
60	1.363	1.883	2.414	2.845	3.126	3.225	3.133	2.859	2.469	2.046
70	1.703	1.583	1.730	2.057	2.468	2.870	3.202	3.425	3.499	3.429

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DATA TABLE 3-2 (Measured by computer. *purpose*: to determine the first excitation potential of argon atom)

V_1=__*2.1V*__; V_2=*2.0V*_____; V_3=__*8.0V*_____;

(The unit of the current in the following table is ____10^{-6} A____)

	0.0	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0
0	0	0	0	0	0	0	0	0	0	0
10	0	0.004	0.020	0.043	0.068	0.089	0.105	0.111	0.106	0.093
20	0.076	0.059	0.046	0.041	0.057	0.106	0.170	0.224	0.251	0.259
30	0.228	0.174	0.116	0.073	0.034	0.062	0.115	0.200	0.284	0.347
40	0.373	0.359	0.304	0.228	0.151	0.097	0.075	0.095	0.164	0.255
50	0.342	0.405	0.435	0.427	0.383	0.312	0.233	0.171	0.135	0.147
60	0.203	0.283	0.364	0.430	0.472	0.485	0.466	0.421	0.360	0.296
70	0.248	0.235	0.261	0.314	0.378	0.442	0.495	0.529	0.540	0.527

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DATA TABLE 3-3 (Measured by computer. *purpose*: to determine the first excitation potential of argon atom)

V_1=___2.2v___; V_2=___2.0V___; V_3=___8.0V___;

(The unit of the current in the following table is ___10^{-7} A___)

	0.0	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0
0	0	0	0	0	0	0	0	0	0	0
10	0.001	0.052	0.185	0.346	0.509	0.463	0.583	0.659	0.671	0.622
20	0.537	0.448	0.366	0.311	0.369	0.663	1.104	1.468	1.716	2.442
30	2.203	1.721	1.179	0.760	0.582	0.701	1.320	2.293	3.264	3.998
40	4.339	4.220	3.638	2.742	1.832	1.194	0.953	1.252	2.162	3.339
50	4.449	5.279	5.701	5.646	5.113	4.206	3.194	2.351	1.885	2.091
60	2.906	4.021	5.141	6.070	6.686	6.904	6.694	6.094	5.253	4.392
70	1.703	1.583	1.730	2.057	2.468	2.870	3.202	3.425	3.499	3.420

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Discussion (More than 150 words, 5 points)

According to the diagram, crests and troughs appear one by one with a regular interval. The interval between each crest turned out to be the first excitation potential of argon atom because the crests mainly formed by the discharge of energy when the atoms jump to another energy level. Additionally, the troughs are formed mainly owing to the loss of energy by electrons due to collisions with atoms or molecules.

Apart from this, the current remained 0 when the V4 was under 10V because the kinetic power would lose when the electric collided with argon atoms. The picoammeter can not indicate the data when the current is too low.

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Conclusions (About 50 words, 2 points)

To sum up, this experiment reveal the essence of atom's first excitation potential. Through the experiment, we also got deeper understanding of Bohr's theory. Our results indicate that the first excitation potential of argon equals to 11.5V, with the relative error of -11%

Score

References (1 points)

1.Hao Fu, Esmond Agurgo Balfour(2017), Introductory Physics Experiments for Undergraduates, *Science Press*, 126-137

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Answers to Questions (6 points)

(1) The values of four voltage have nothing to do with the first excitation potential of the argon atom because the first excitation potential of the argon atom remains constant at approximately 13.1V.

(2) The Franck-Hertz experiment directly proved that atoms have the "fully determined, discrete energy state" envisaged by Bohr. This finding was the first decisive proof of Bohr's model of atomic quantization.

(3) Once atoms are excited, their electrons absorb energy and move to higher energy levels. This excited statis unstable, so the excited atoms eventually return to their ground state.

According to the energy-wavelength relationship given by the formula $E=hc/\lambda$, where E is the energy, h is Planck's constant, c is the speed of light, and λ is the wavelength, λ=94.9nm, use ultraviolet detector to detect

(4)When the voltage V1 increases, the current IP will also increase. Consequently, when V1 reaches a sufficiently high level, the initial speed and energy of the electrons will be elevated to the point where they can more readily surpass the obstacle posed by V3.

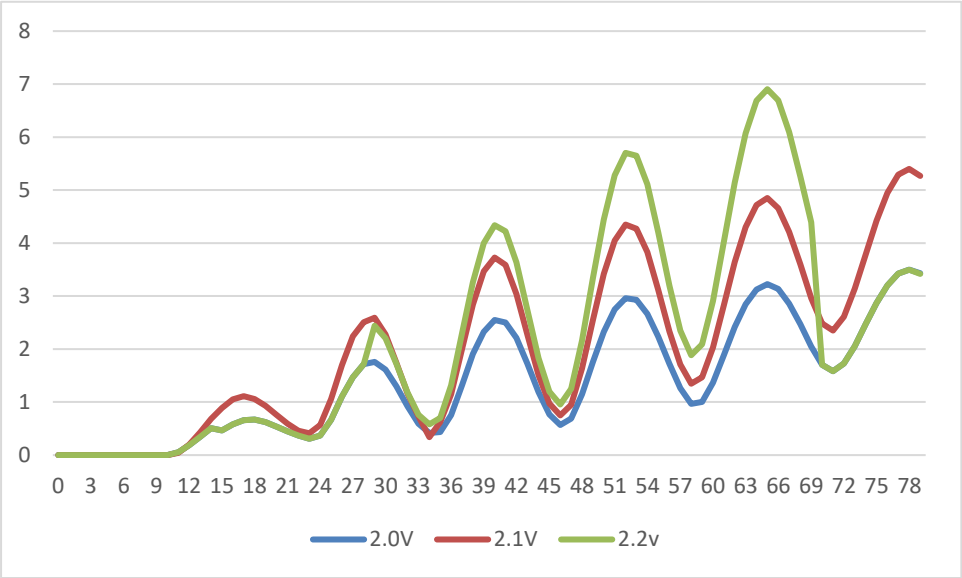
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Appendix

Score

(Calculations, 15 points)

6. Calculations



Compute the first excitation potential, U_0, from the graph.

(1) Choose one trace and read the horizontal coordinates of the crests. Put the voltages in the following table.

Crest	1	2	3	4	5	6
Voltages, U (V)	18	29	40	52	65	78

(2) Compute the difference of the voltages

$U_{01}=U_2-U_1=(29-18)V=11V$
$U_{02}=U_3-U_1=(40-18)V=22V$
$U_{03}=U_4-U_1=(52-18)V=34V$
$U_{04}=U_5-U_1=(65-18)V=47V$
$U_{05}=U_6-U_1=(78-18)V=60V$

(3) Compute to the first excitation potential U_0.

$$U_0=\frac{U_{01}+U_{02}+U_{03}+U_{04}+U_{05}}{1+2+3+4+5}=\frac{11V+22V+34V+47V+60V}{15}=11.6V$$

(4) The accepted value of the first excitation potential of argon is 13.1 V. Compute the relative error of your measurement.

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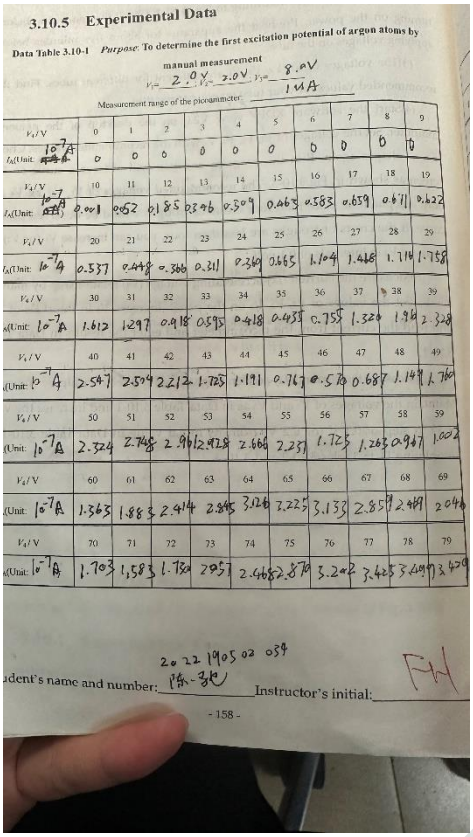
the relative error of you measurement=$\frac{11.6V-13,1V}{13.1V}$=-11%

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Appendix

(Scanned data sheets)



3.10.5 Experimental Data

Data Table 3.10-1 *Purpose:* To determine the first excitation potential of argon atoms by manual measurement

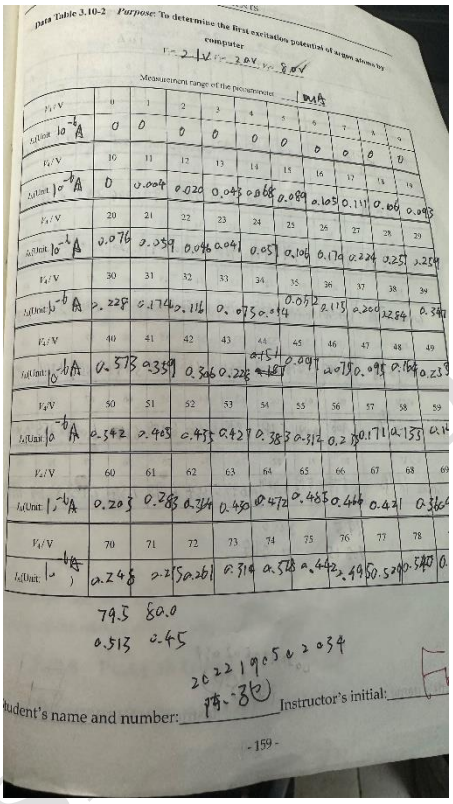
Measurement range of the picoammeter: 1 μA

V_A/V	0	1	2	3	4	5	6	7	8	9
I_A(Unit: 10^{-7}A)	0	0	0	0	0	0	0	0	0	0
V_A/V	10	11	12	13	14	15	16	17	18	19
I_A(Unit: 10^{-7}A)	0.001	0.052	0.185	0.396	0.509	0.463	0.583	0.659	0.671	0.622
V_A/V	20	21	22	23	24	25	26	27	28	29
I_A(Unit: 10^{-7}A)	0.537	0.448	0.366	0.311	0.369	0.565	1.104	1.468	1.714	1.758
V_A/V	30	31	32	33	34	35	36	37	38	39
I_A(Unit: 10^{-7}A)	1.612	1.297	0.918	0.595	0.418	0.455	0.755	1.320	1.96	2.328
V_A/V	40	41	42	43	44	45	46	47	48	49
I_A(Unit: 10^{-7}A)	2.547	2.594	2.212	1.725	1.191	0.767	0.579	0.687	1.149	1.760
V_A/V	50	51	52	53	54	55	56	57	58	59
I_A(Unit: 10^{-7}A)	2.324	2.748	2.996	2.928	2.666	2.285	1.725	1.263	0.967	1.002
V_A/V	60	61	62	63	64	65	66	67	68	69
I_A(Unit: 10^{-7}A)	1.363	1.883	2.414	2.845	3.126	3.225	3.133	2.859	2.488	2.048
V_A/V	70	71	72	73	74	75	76	77	78	79
I_A(Unit: 10^{-7}A)	1.703	1.583	1.752	2.057	2.468	2.876	3.242	3.485	3.499	3.479

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Data Table 3.10-2 *Purpose:* To determine the first excitation potential of argon atoms by computer

Measurement range of the picoammeter: 1 μA

V_A/V	0	1	2	3	4	5	6	7	8	9
I_A(Unit: 10^{-6}A)	0	0	0	0	0	0	0	0	0	0
V_A/V	10	11	12	13	14	15	16	17	18	19
I_A(Unit: 10^{-7}A)	0	0.004	0.020	0.043	0.068	0.089	0.103	0.111	0.106	0.093
V_A/V	20	21	22	23	24	25	26	27	28	29
I_A(Unit: 10^{-7}A)	0.076	0.059	0.046	0.041	0.05	0.106	0.174	0.224	0.251	0.254
V_A/V	30	31	32	33	34	35	36	37	38	39
I_A(Unit: 10^{-6}A)	0.228	0.174	0.116	0.073	0.044	0.052	0.115	0.200	0.284	0.347
V_A/V	40	41	42	43	44	45	46	47	48	49
I_A(Unit: 10^{-6}A)	0.378	0.359	0.306	0.226	0.154	0.094	0.078	0.095	0.164	0.238
V_A/V	50	51	52	53	54	55	56	57	58	59
I_A(Unit: 10^{-6}A)	0.342	0.403	0.435	0.427	0.383	0.314	0.235	0.171	0.133	0.124
V_A/V	60	61	62	63	64	65	66	67	68	69
I_A(Unit: 10^{-6}A)	0.203	0.288	0.374	0.450	0.472	0.483	0.444	0.421	0.386	0.364
V_A/V	70	71	72	73	74	75	76	77	78	
I_A(Unit: 10^{-6}A)	0.248	0.215	0.261	0.314	0.428	0.442	0.490	0.520	0.540	0

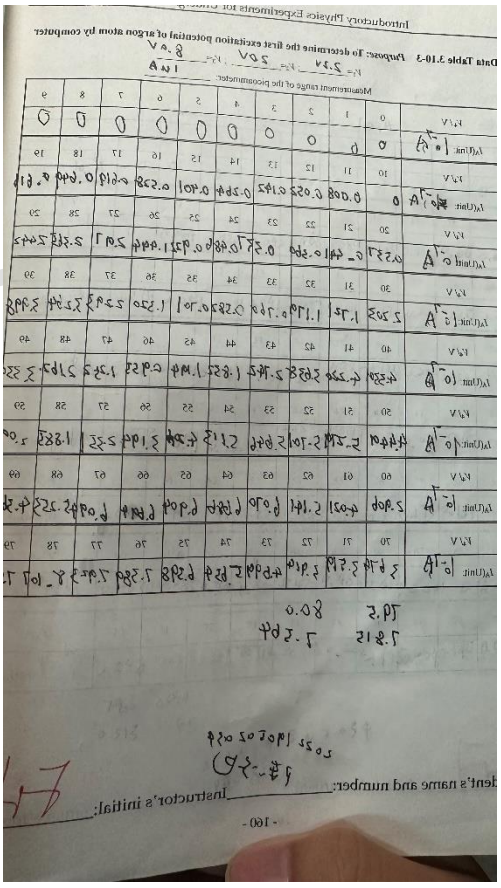
79.5 80.0

0.513 0.45

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Data Table 3.10-3 *Purpose:* To determine the first excitation potential of argon atom by computer

Measurement range of the picoammeter: 1 μA

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